Task 2: How Kenyan maternal/child indicators (stunting, skilled birth attendance, immunisation) vary by region and wealth quintile - Descriptive and exploratory analysis

31 July 2026

Loading the KDHS Data 2022 (KR Record)

```
rm(list = ls())

# Packages
pacman::p_load(
  tidyverse, haven, labelled, epiDisplay, survey,
  gtsummary, extrafont, flextable, rKenyaCensus, sf, rineq)

# Data
df <- read_dta("../Mini project/Data/KEKR8BDT/KEKR8BFL.DTA")
```

```
plot_pointrange <-
    function(data, x_var, estimate, lower, upper,
    reference_line = NULL, y_limits = c(5, 50),
    y_breaks = seq(5, 50, 5),
    y_label = "Weighted prevalence (%)") {

  x_var <- rlang::ensym(x_var)
  estimate <- rlang::ensym(estimate)
  lower <- rlang::ensym(lower)
  upper <- rlang::ensym(upper)

  p <- data |>
    ggplot(
      aes(
        x = forcats::fct_reorder(!!x_var, !!estimate, .desc = TRUE),
        y = !!estimate)) +
    geom_pointrange(
      aes(
        ymin = !!lower,
        ymax = !!upper),
      size = .3,
      color = "gray30") +
    scale_y_continuous(
      limits = y_limits,
      breaks = y_breaks) +
    labs(
      x = NULL,
      y = y_label) +
```

```
    theme_bw(
      base_family = "Candara",
      base_size = 14) +
    theme(
      axis.text = element_text(color = "black"),
      axis.text.x = element_text(angle = 90, vjust = .5))

  if (!is.null(reference_line)) {
    p <- p +
      geom_hline(
        yintercept = reference_line,
        lty = 2,
        color = "blue")
  }

  return(p)
}
```

```
plot_map <- function(
    data, fill, breaks, labels, legend_title, subtitle, palette = "YlGnBu",
direction = 1) {

  fill <- rlang::ensym(fill)

  data |>
    mutate(
      fill_cat = cut(
        !!fill,
        breaks = breaks,
        labels = labels,
        right = FALSE,
        include.lowest = TRUE)) |>
    ggplot() +
    geom_sf(
      aes(fill = fill_cat),
      color = "gray30") +
    scale_fill_brewer(
      palette = palette,
      direction = direction) +
    labs(
      fill = legend_title,
      subtitle = subtitle) +
    theme_minimal(base_family = "Candara", base_size = 14) +
    theme(
      axis.text = element_blank(),
      plot.subtitle = element_text(hjust = .6, vjust = -8),
      legend.background = element_rect(color = "gray70"),
      legend.title = element_text(size = 10.5),
      legend.text = element_text(size = 10.5),
      legend.position = c(0.15, 0.16),
```

```
      panel.grid = element_blank()
    )
}
```

Data preparation

```
## Restrict the analysis to the youngest alive child, living with the responded
df <- df |>
    mutate(
        age_months = v008 - b3) |>
    filter(
        b5 == 1, # child is alive
        b9 == 0, # child lives with the respondent
        !is.na(v003) # mother's line number must be available
        ) |>
    group_by(
        v001,
        v002,
        v003 # grouping by cluster, household, and mothers line number
    ) |>
    mutate(
        bidxr =
            rank(bidx, ties.method = "first")) |>
    filter(
        bidxr == 1 # Keeping the youngest child
        ) |>
    ungroup()

df <- df |>
    mutate(
        wt = v005/1000000, # sample weight
        wealth_cont = v191/100000,
        wealth_q = v190
    )
```

The datasets includes a total of 13528 children from 1689 clusters across Kenya.

The ouctomes of interest

Stunting

National weighted n and prevalence of stunting

```
df_stunting <- df |>
    mutate(
    hw70 = ifelse( #hw70 is the height-for-age zscore and should be divided by 100
        hw70 >
         1000, NA, hw70),
    haz = hw70/100,

    stunted = ifelse(haz < -2, "Stunted", "Not stunted"),
    stun = ifelse(stunted == "Stunted", 1, 0)) |>
    filter(
        age_months >= 12 & age_months < 36,
        !is.na(stunted))

# Saving a copy of the data for later MBG analysis
write_dta(
    df_stunting,
    "./Data/Derived_data/df_stunting.dta")

df_stunting |>
    ggplot(
        aes(x = haz)) +
            geom_histogram(color = "white") +
            labs(
                x = "Height-for-age z-score (HAZ scores)",
                y = "Frequency") +
    theme_bw(base_family = "Candara", base_size = 14) +
    theme(panel.grid = element_blank())
```

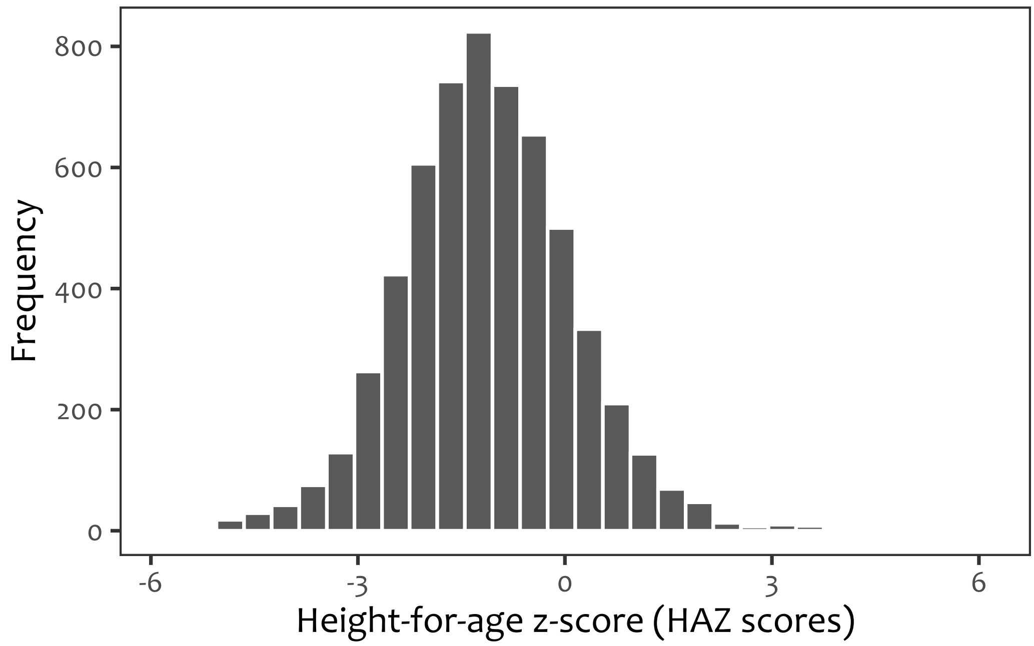

Figure 1: Histogram of HAZ scores

```
# Survey setting
stunting_survey <- svydesign(
  id = ~v021, # primary sampling unit
  strata = ~v022, # strata
  weights = ~wt,
  data = df_stunting,
  nest = TRUE)

options(survey.lonely.psu = "adjust") # adjust for case where there may be 1 psu
in a strata

# Weighted prevalence of stunting
tbl <- tbl_svysummary(
  stunting_survey,
  include = stunted,
  statistic = all_categorical() ~ "{n} ({p})",
  digits = all_categorical() ~ c(1, 1),
  missing = "ifany",
  missing_text = "Missing",
  label = stunted ~ "Stunting status")
```

Stunting was defined as height-for-age z-score (HAZ) < −2 based on the WHO 2006 growth reference standards. The analysis included a total of 5883. According to the WHO guidelines, HAZ scores range from −6 to 6 and this was ascertained for the data as shown in Figure 1. The data was then survey-set to account for the two-stage sampling design of DHS. The weighted n (prevalence %) of stunting among the 12-35 months old children was `tbl$table_body$stat_0[[3]]`.

Does stunting vary by region (subnational levels/counties)?

```
county_stunting <- svyby(
  ~ I(stunted == "Stunted"),
  by = ~v024,
  design = stunting_survey,
  FUN = svyciprop,
  vartype = "ci",
  na.rm = TRUE)

stunting_county <- county_stunting |>
    mutate(
        county = v024,
        prevalence = `I(stunted == "Stunted")`,
        lb = ci_l,
        ub = ci_u) |>
    dplyr::select(county, prevalence, lb, ub)


stunting_county <- stunting_county |>
    mutate(
        name = names(attr(county, "labels")) |>
            str_to_title() |>
            str_replace_all("[[:punct:]]", " "))


# Fetching Kenya counties shapefiles
kenya_shp <- rKenyaCensus::KenyaCounties_SHP |>
    janitor::clean_names() |>
    st_as_sf() |>
    dplyr::select(county) |>
    mutate(
        county = str_to_title(county) |>
            str_replace_all("[[:punct:]]", " "))

kenya_shp[kenya_shp$county == "Nairobi City", "county"] <- "Nairobi"

# Merge the prevalence data to the shp file data
merged_stunting <- kenya_shp |>
    left_join(
        stunting_county, by = c("county" = "name")) |>
    mutate(
        across(
            c(prevalence, lb, ub), ~round(., 3)*100))

# Plotting the prevalence across the counties
plot_pointrange(
    merged_stunting, x_var = "county", estimate = "prevalence",
    lower = lb, upper = ub,
    reference_line =
        as.numeric(
            str_extract(tbl$table_body$stat_0[[3]],
```

```
            "(?<=\\()[0-9.]+(?=\\))")),
    y_limits = c(0, 60),
    y_breaks = seq(0, 60, 5),
    y_label = "Weighted prevalence (%)") +
  annotate(
    geom = "text",
    label = paste0("The blue line is the national average = ",
    as.numeric(
            str_extract(tbl$table_body$stat_0[[3]],
             "(?<=\\()[0-9.]+(?=\\))")), "%"),
    x = 25,
    y = 55) +
    theme(
      axis.text.x = element_text(size = 12),
      panel.grid = element_line(color = "gray96"))

ggsave(
  "./Plots/Stunting_forest_plot.png",
  width = 10,
  height = 6,
  dpi = 500,
  bg = "white")
```

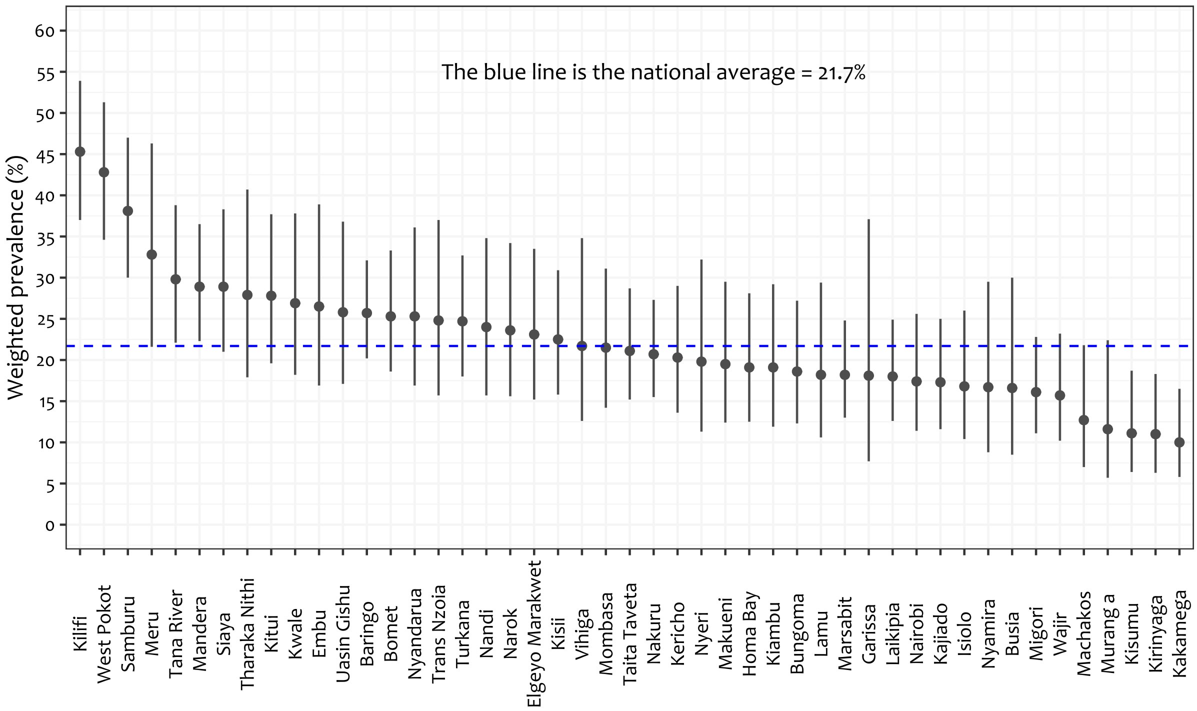

Figure 2: Weighted prevalence of stunting among children 12-35 months at the subnationa levels. The blue line represents the national weighted prevalence

Figure 2 shows the weighted prevalence of stunting among children 12-35 months at the subnational levels with the blue dotted line showing the weighted national prevalence.

```r
## Prevalence maps
plot_map(
    merged_stunting, fill = prevalence,
    breaks = c(0, 10, 20, 30, 40, 50),
    labels = c("0–10", "11–20", "21–30", "31–40", "41–50"),
    legend_title = "Prevalence (%)", subtitle = NULL,
    palette = 7, direction = 1) +
  theme(
    legend.text = element_text(size = 9),
    legend.background = element_blank(),
    legend.position = c(.18, .12),
    plot.background = element_rect(fill = "gray97"))

ggsave(
  "./Plots/Stunting_map.png",
  width = 8,
  height = 6,
  dpi = 500,
  bg = "white")
```

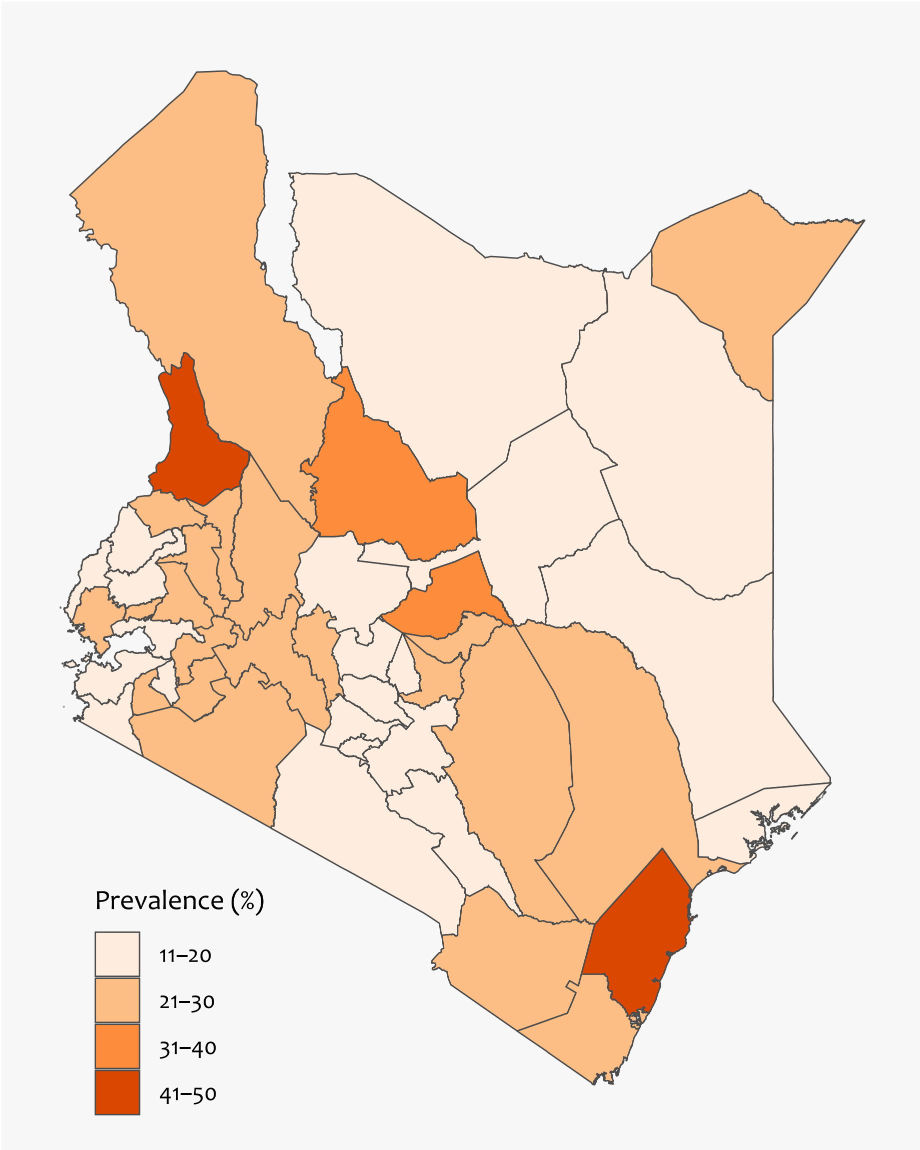

Figure 3: Weighted prevalence of stunting among children 6-59 months at the subnationa levels

Figure 3 shows the weighted prevalence of stunting among children 6-59 months at the subnational levels. Darker shades shows increasing prevalence

Does stunting vary by wealth status?

Weighted concentration curves

```
cc_df <- df_stunting |>
    filter(!is.na(stun), !is.na(wealth_cont), !is.na(wt)) |>
    arrange(wealth_cont) |>
    mutate(
        cum_prop = cumsum(wt) / sum(wt),
        cum_stunted = cumsum(stun*wt) / sum(stun*wt))


cc_df <- df_stunting |>
    filter(!is.na(stun), !is.na(wealth_cont), !is.na(wt)) |>
```

```r
    arrange(wealth_cont) |>
    mutate(
    pop_rank = cumsum(wt) - wt / 2, # midpoint rank within own weight mass
    pop_rank = pop_rank / sum(wt)) |>
  group_by(wealth_cont) |>   # collapse exact ties
  mutate(pop_rank = mean(pop_rank)) |> # give tied group its average rank
  ungroup() |>
  arrange(pop_rank) |>
  mutate(
        cum_prop = cumsum(wt) / sum(wt),
        cum_stunted = cumsum(stun*wt) / sum(stun*wt))

cc_plot_data <- bind_rows(
  data.frame(cum_prop = 0, cum_stunted = 0),
  cc_df |> dplyr::select(cum_prop, cum_stunted)) # Adding (0, 0) for plotting
```

```r
## Concentration index
ci_result <- ci(
  ineqvar = df_stunting$wealth_cont,
  outcome = df_stunting$stun,
  weights = df_stunting$wt,
  type    = "CIw")

cc_plot_data |>
  ggplot(aes(x = cum_prop, y = cum_stunted)) +
  geom_line(color = "navyblue", linewidth = 1) +
  geom_abline(
    slope = 1, intercept = 0,
     color = "#813134") +
  annotate(
    "text",
    x = 0.05, y = 0.95,
    label = paste0(
      "CI (95% CI) = ",
      round(ci_result$concentration_index, 2),
      " (",
      round(confint(ci_result)[[1]], 2), ",", " ",
      round(confint(ci_result)[[2]], 2), ")"),
    family = "Candara",
    hjust = 0, vjust = 1,
    size = 4.0) +
  labs(
    x = "Cumulative % of children, ranked by wealth (poorest \u2192 richest)",
    y = "Cumulative % of stunted children") +
  theme_classic(base_family = "Candara", base_size = 13) +
  theme(
    panel.grid = element_blank())

ggsave(
  "./Plots/Stunting_concentration_curve.png",
```

```
  width = 7,
  height = 5,
  dpi = 500,
  bg = "white")
```

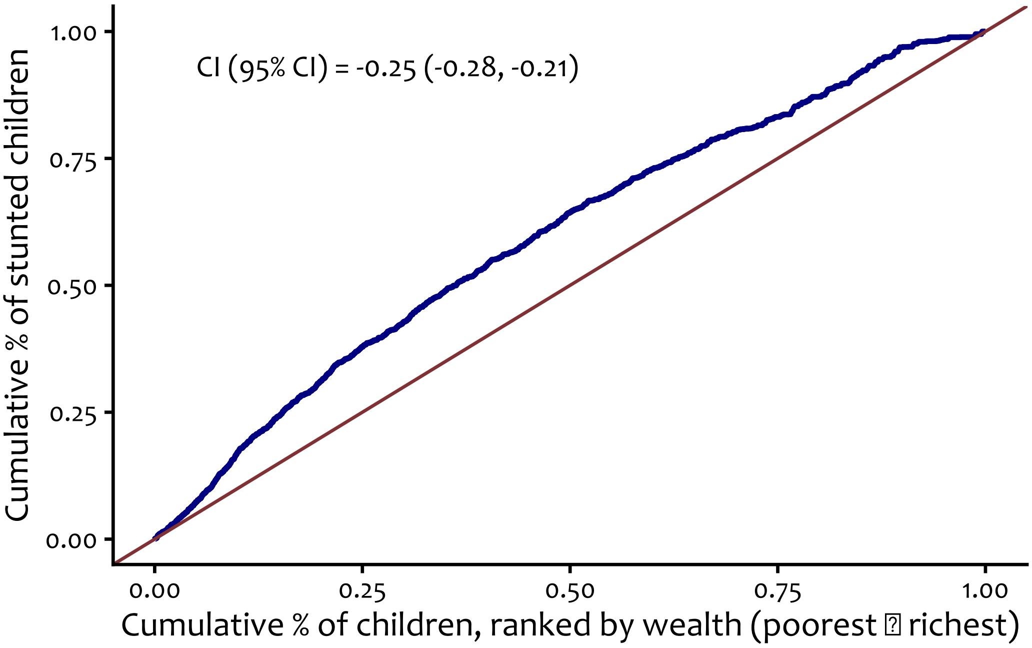

Figure 4: Concentration curves showing equity/inequity in the distribution of stunting by wealth status

Immunisation

Vaccination analysis was restricted to children aged 12-35 months. Full vaccination was defined differently for these two groups of children according to the national vaccination schedule.

Distribution of full vaccination at the national and subnational levels, children 12-35 months

Full vaccination was defined as having taken one dose of BCG vaccine,oral polio vaccine (OPV, birth dose), three doses of OPV and one dose of inactivated polio vaccine (IPV), three doses of DPT-HepB-Hib, three doses of pneumococcal conjugate vaccine (PCV), two doses of Rotavirus vaccine (RV), and one dose of measles rubella (MR) among children 12-23 months, and additionally having received thes second dose of measles rubella of children 24-35 months.

```
# Defining fully vaccinated children 12-23 years
df_immunn_12_23 <- df |>
    filter(
        age_months >= 12 & age_months < 24) |>
    mutate(
        fully_vaccinated =
            ifelse(

            (h2 %in% c(1, 2, 3) & # one dose of BCG vaccine

            h0 %in% c(1, 2, 3) & # Oral polio vaccine (birth dose)
            h4 %in% c(1, 2, 3) & # Oral polio vaccine 1
            h6 %in% c(1, 2, 3) & # Oral polio vaccine 2
            h8 %in% c(1, 2, 3) & # Oral polio vaccine 3
            h60 %in% c(1, 2, 3) & # one dose of inactivated polio vaccine

            h3 %in% c(1, 2, 3) & # Dose 1 DPT-HepB-Hib
            h5 %in% c(1, 2, 3) & # Does 2 DPT-HepB-Hib
            h7 %in% c(1, 2, 3) & # Dose 3 DPT-HepB-Hib

            h54 %in% c(1, 2, 3) & # Pneumococcal conjugate vaccine 1
            h55 %in% c(1, 2, 3) & # Pneumococcal conjugate vaccine 2
            h56 %in% c(1, 2, 3) & # Pneumococcal conjugate vaccine 3

            h57 %in% c(1, 2, 3) & # Rotavirus vaccine 1
            h58 %in% c(1, 2, 3) & # Rotavirus vaccine 2

            h9 %in% c(1, 2, 3)  # Measles rubella 1
            ), "Fully vaccinated", "Not fully vaccinated"),
        fully_vacc = ifelse(
            fully_vaccinated == "Fully vaccinated", 0, 1)
    )
```

```
# Defining fully vaccinated children 24-35 years
df_immunn_24_35 <- df |>
    filter(
        age_months >= 24 & age_months < 36) |>
    mutate(
        fully_vaccinated =
            ifelse(

            (h2 %in% c(1, 2, 3) & # one dose of BCG vaccine

            h0 %in% c(1, 2, 3) & # Oral polio vaccine (birth dose)
            h4 %in% c(1, 2, 3) & # Oral polio vaccine 1
            h6 %in% c(1, 2, 3) & # Oral polio vaccine 2
            h8 %in% c(1, 2, 3) & # Oral polio vaccine 3
            h60 %in% c(1, 2, 3) & # one dose of inactivated polio vaccine

            h3 %in% c(1, 2, 3) & # Dose 1 DPT-HepB-Hib
            h5 %in% c(1, 2, 3) & # Does 2 DPT-HepB-Hib
            h7 %in% c(1, 2, 3) & # Dose 3 DPT-HepB-Hib

            h54 %in% c(1, 2, 3) & # Pneumococcal conjugate vaccine 1
            h55 %in% c(1, 2, 3) & # Pneumococcal conjugate vaccine 2
            h56 %in% c(1, 2, 3) & # Pneumococcal conjugate vaccine 3

            h57 %in% c(1, 2, 3) & # Rotavirus vaccine 1
            h58 %in% c(1, 2, 3) & # Rotavirus vaccine 2

            h9 %in% c(1, 2, 3) &  # Measles rubella 1
            h9a %in% c(1, 2, 3)   # Measles rubella 2
            ), "Fully vaccinated", "Not fully vaccinated"),
        fully_vacc = ifelse(
            fully_vaccinated == "Fully vaccinated", 0, 1)
    )
```

```
# Merging the two sets
df_immunn <- bind_rows(
  df_immunn_12_23,
  df_immunn_24_35)
```

Survey setting step

```
immunn_survey <- svydesign(
  id = ~v021, # primary sampling unit
  strata = ~v022, # strata
  weights = ~wt,
  data = df_immunn,
  nest = TRUE)

options(survey.lonely.psu = "adjust") # adjust for case where there may be 1 psu
in a strata
```

Weighted prevalence of full vaccination - national level

```
tbl_immun <- tbl_svysummary(
  immunn_survey,
  include = fully_vaccinated,
  statistic = all_categorical() ~ "{n} ({p})",
  digits = all_categorical() ~ c(1, 1),
  missing = "ifany",
  missing_text = "Missing",
  label = fully_vaccinated ~ "Vaccination status")
```

Weighted prevalence of full vaccination - subnational levels

```
county_immunn <- svyby(
  ~ I(fully_vaccinated == "Fully vaccinated"),
  by = ~v024,
  design = immunn_survey,
  FUN = svyciprop,
  vartype = "ci",
  na.rm = TRUE)

immunn_county <- county_immunn |>
    mutate(
        county = v024,
        prevalence = `I(fully_vaccinated == "Fully vaccinated")`,
        lb = ci_l,
        ub = ci_u,
        group = rep("12-23 months")) |>
    dplyr::select(county, group, prevalence, lb, ub)


immunn_county <- immunn_county |>
    mutate(
        name = names(attr(county, "labels")) |>
            str_to_title() |>
            str_replace_all("[[:punct:]]", " "))


# Merge the prevalence data to the shp file data
merged_immunn <- kenya_shp |>
    left_join(
        immunn_county, by = c("county" = "name")) |>
    mutate(
        across(
            c(prevalence, lb, ub), ~round(., 3)*100))

# Plotting
plot_pointrange(
    merged_immunn, x_var = "county", estimate = "prevalence",
    lower = lb, upper = ub,
    reference_line =
```

```r
        as.numeric(
            str_extract(tbl_immun$table_body$stat_0[[3]],
             "(?<=\\()[0-9.]+(?=\\))")),
    y_limits = c(0, 100),
    y_breaks = seq(0, 100, 10),
    y_label = "Weighted prevalence (%)") +
  annotate(
    geom = "text",
    label = paste0("The blue line is the national average = ",
    as.numeric(
            str_extract(tbl_immun$table_body$stat_0[[3]],
             "(?<=\\()[0-9.]+(?=\\))")), "%"),
    x = 25,
    y = 90) +
    theme(
      axis.text.x = element_text(size = 12),
      panel.grid = element_line(color = "gray96"))

ggsave(
  "./Plots/Vaccination_forest_plot.png",
  width = 10,
  height = 6,
  dpi = 500,
  bg = "white")
```

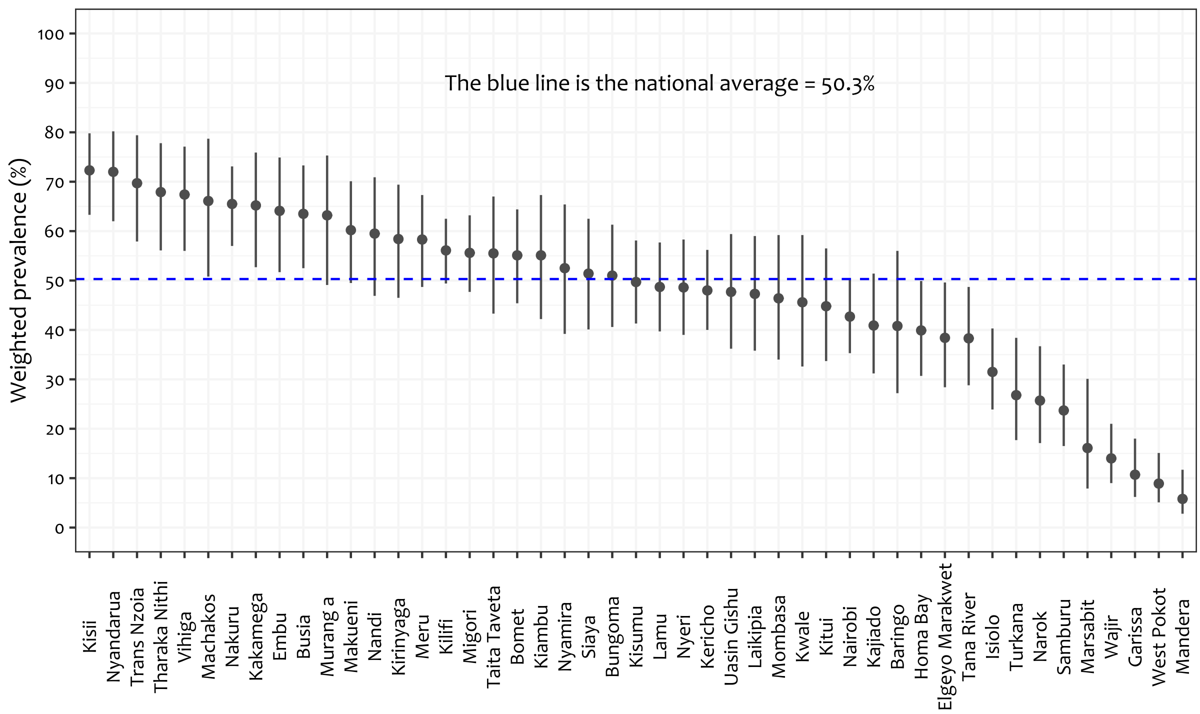

Figure 5: Weighted prevalence of full vacciantion among children 12-35 months at the subnationa levels. The blue line represents the national weighted prevalence

Spatial distribution of full vacciantion among children 12-35 months at the subnationa level

```r
plot_map(
  merged_immunn,
  fill = prevalence,
  breaks = c(0, 20, 40, 60, 80, 100),
  labels = c("<20", "20–39.9", "40–59.9", "60–79.9", "80–100"),
  legend_title = "Prevalence (%)",
  subtitle = NULL,
  palette = 7,
  direction = -1) +
  theme(
    legend.text = element_text(size = 9),
    legend.background = element_blank(),
    legend.position = c(.18, .12),
    plot.background = element_rect(fill = "gray97"))

ggsave(
  "./Plots/Vaccination_map.png",
  width = 8,
  height = 6,
  dpi = 500,
  bg = "white")
```

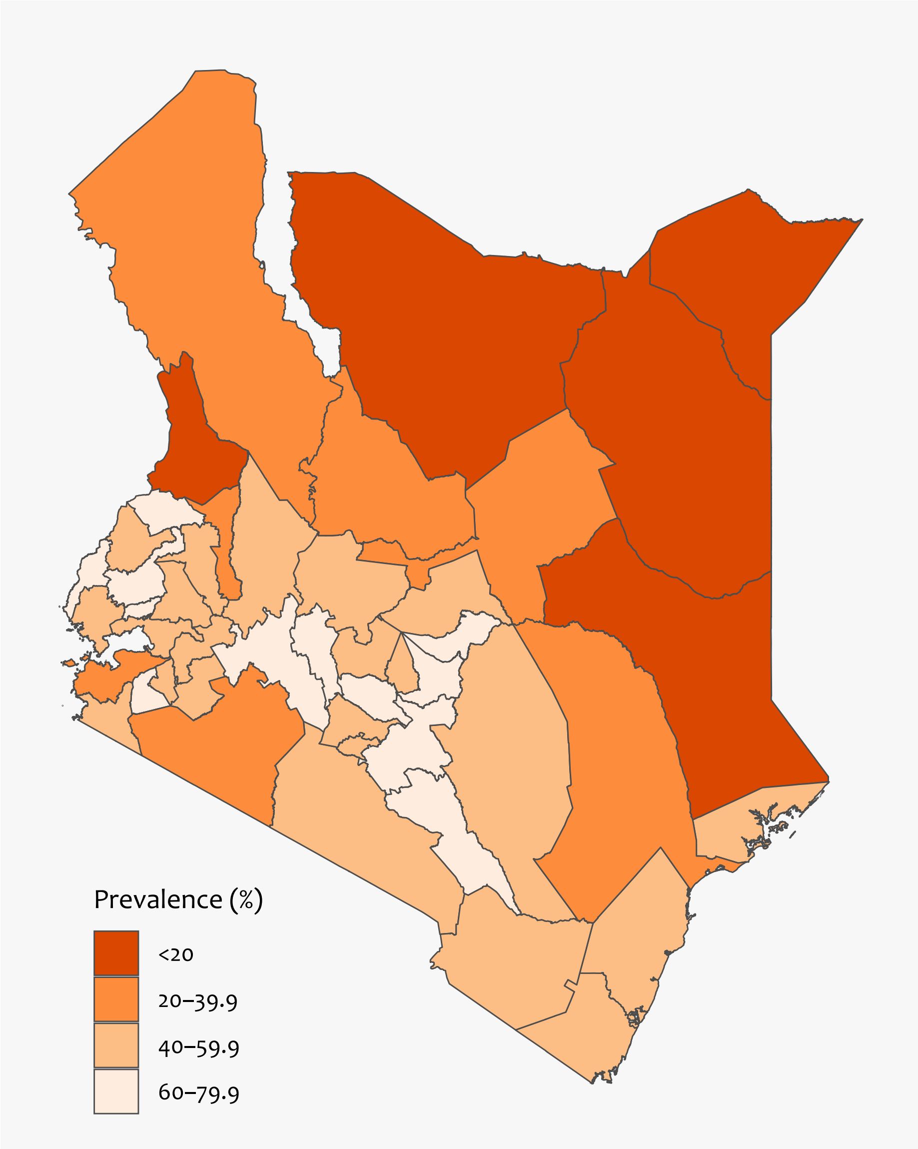

Figure 6: Spatial distribution of full vacciantion among children 12-35 months at the subnationa level

Concentration curve for full vaccination by cumulative percentage of the population ranked by wealth index

Data preparation step

```
cc_df_immunn <- df_immunn |>
    filter(!is.na(fully_vacc), !is.na(wealth_cont), !is.na(wt)) |>
    arrange(wealth_cont) |>
    mutate(
    pop_rank = cumsum(wt) - wt / 2, # midpoint rank within own weight mass
    pop_rank = pop_rank / sum(wt)) |>
  group_by(wealth_cont) |>   # collapse exact ties
  mutate(pop_rank = mean(pop_rank)) |> # give tied group its average rank
  ungroup() |>
  arrange(pop_rank) |>
  mutate(
        cum_prop = cumsum(wt) / sum(wt),
```

```
        cum_full_vacc = cumsum(fully_vacc*wt) / sum(fully_vacc*wt))


cc_plot_data_immunn <- bind_rows(
  data.frame(cum_prop = 0, cum_full_vacc = 0),
  cc_df_immunn |> dplyr::select(cum_prop, cum_full_vacc)) # Adding (0, 0) for
plotting
```

Generating concentration index and plotting

```
## Concentration index
ci_result_immunn <- ci(
  ineqvar = df_immunn_12_23$wealth_cont,
  outcome = df_immunn_12_23$fully_vacc,
  weights = df_immunn_12_23$wt,
  type    = "CIw")

cc_plot_data_immunn |>
  ggplot(aes(x = cum_prop, y = cum_full_vacc)) +
  geom_line(color = "darkgreen", linewidth = 1) +
  geom_abline(
    slope = 1, intercept = 0,
     color = "#813134") +
  annotate(
    "text",
    x = 0.05, y = 0.95,
    label = paste0(
      "Concentration index (95% CI) = ",
      round(ci_result_immunn$concentration_index, 2),
      " (",
      round(confint(ci_result_immunn)[[1]], 2), ",", " ",
      round(confint(ci_result_immunn)[[2]], 2), ")"),
    family = "Candara",
    hjust = 0, vjust = 1,
    size = 4.0) +
  labs(
    x = "Cumulative % of children, ranked by wealth (poorest \u2192 richest)",
    y = "Cumulative % of non fully vaccinated children",
    subtitle = NULL) +
  theme_classic(base_family = "Candara", base_size = 13) +
  theme(
    panel.grid = element_blank())

  ggsave(
  "./Plots/Vaccination_concentration_curve.png",
  width = 7,
  height = 5,
  dpi = 500,
  bg = "white")
```

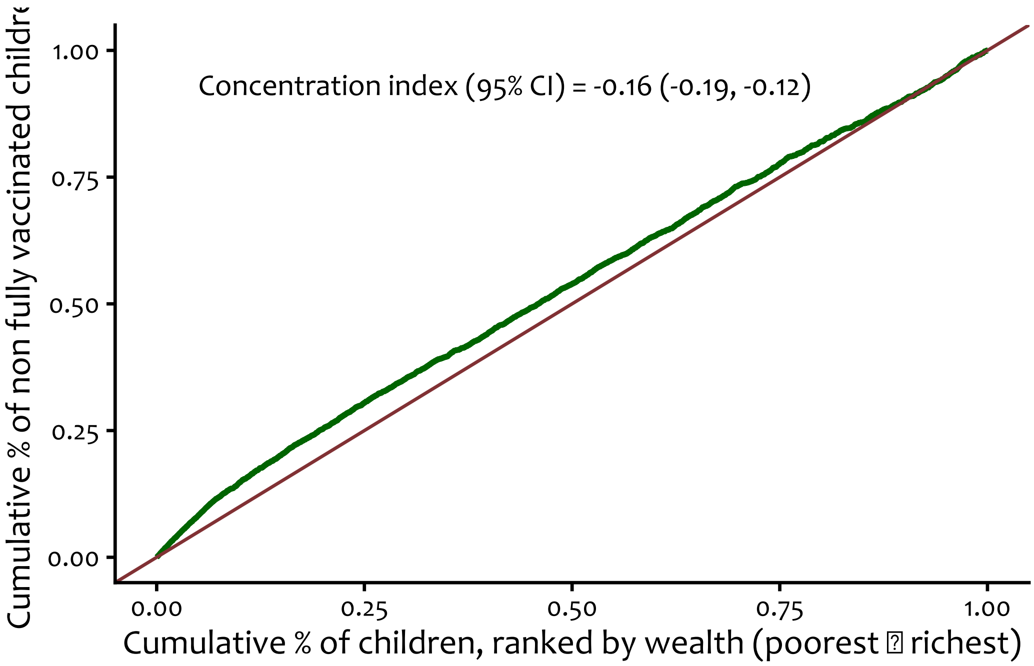

Figure 7: Concentration curves showing equity/inequity in the distribution of full vaccination by wealth status

Skilled birth attendance

Skilled birth attendance (SBA) analysis was restricted to children aged 12–35 months. In the 2022 KDHS, delivery-assistance (SBA) questions were asked only about a mother's most recent birth, and only if that child was aged 0–35 months at the time of the interview.

SBA was defined as a delivery assisted by a doctor, nurse, or midwife/clinical officer, in line with the standard DHS/WHO definition of a trained health professional able to manage normal deliveries and identify, manage, and refer obstetric complications (KDHS 2022 variables m3a, m3b). Traditional birth attendants, relatives, and other unskilled providers were not classified as skilled.

Distribution of skilled birth attendance at the national and subnational levels, children 12-35 months

```
df_sba <- df |>
  filter(
    !is.na(m3a), !is.na(m3b),
    age_months >= 12 & age_months < 36) |>
  mutate(
    sba = if_else(m3a == 1 | m3b == 1, "sba", "non sba"),
    sba_num = if_else(m3a == 1 | m3b == 1, 1, 0))
```

Survey setting step

```
df_sba_survey <- svydesign(
  id = ~v021, # primary sampling unit
  strata = ~v022, # strata
  weights = ~wt,
  data = df_sba,
  nest = TRUE)

options(survey.lonely.psu = "adjust") # adjust for case where there may be 1 psu
in a strata
```

Weighted prevalence of sba - national level

```
tbl_sba <- tbl_svysummary(
  df_sba_survey,
  include = sba,
  statistic = all_categorical() ~ "{n} ({p})",
  digits = all_categorical() ~ c(1, 1),
  missing = "ifany",
  missing_text = "Missing",
  label = sba ~ "Skilled birth attendance")
```

Weighted prevalence of full vaccination - subnational levels

```r
county_sba <- svyby(
  ~ I(sba == "sba"),
  by = ~v024,
  design = df_sba_survey,
  FUN = svyciprop,
  vartype = "ci",
  na.rm = TRUE)

sba_county <- county_sba |>
    mutate(
        county = v024,
        prevalence = `I(sba == "sba")`,
        lb = ci_l,
        ub = ci_u) |>
    dplyr::select(county, prevalence, lb, ub)


sba_county <- sba_county |>
    mutate(
        name = names(attr(county, "labels")) |>
            str_to_title() |>
            str_replace_all("[[:punct:]]", " "))


# Merge the prevalence data to the shp file data
merged_sba <- kenya_shp |>
    left_join(
        sba_county, by = c("county" = "name")) |>
    mutate(
        across(
            c(prevalence, lb, ub), ~round(., 3)*100))

# Plotting
plot_pointrange(
    merged_sba, x_var = "county", estimate = "prevalence",
    lower = lb, upper = ub,
    reference_line =
        as.numeric(
            str_extract(tbl_sba$table_body$stat_0[[3]],
             "(?<=\\()[0-9.]+(?=\\))")),
    y_limits = c(30, 100),
    y_breaks = seq(30, 100, 10),
    y_label = "Weighted prevalence (%)") +
    annotate(
    geom = "text",
    label = paste0("The blue line is the national average = ",
    as.numeric(
            str_extract(tbl_sba$table_body$stat_0[[3]],
             "(?<=\\()[0-9.]+(?=\\))")), "%"),
    x = 15,
```

```r
    y = 60) +
    theme(
      axis.text.x = element_text(size = 12),
      panel.grid = element_line(color = "gray96"))

ggsave(
  "./Plots/Sba_forest_plot.png",
  width = 10,
  height = 6,
  dpi = 500,
  bg = "white")
```

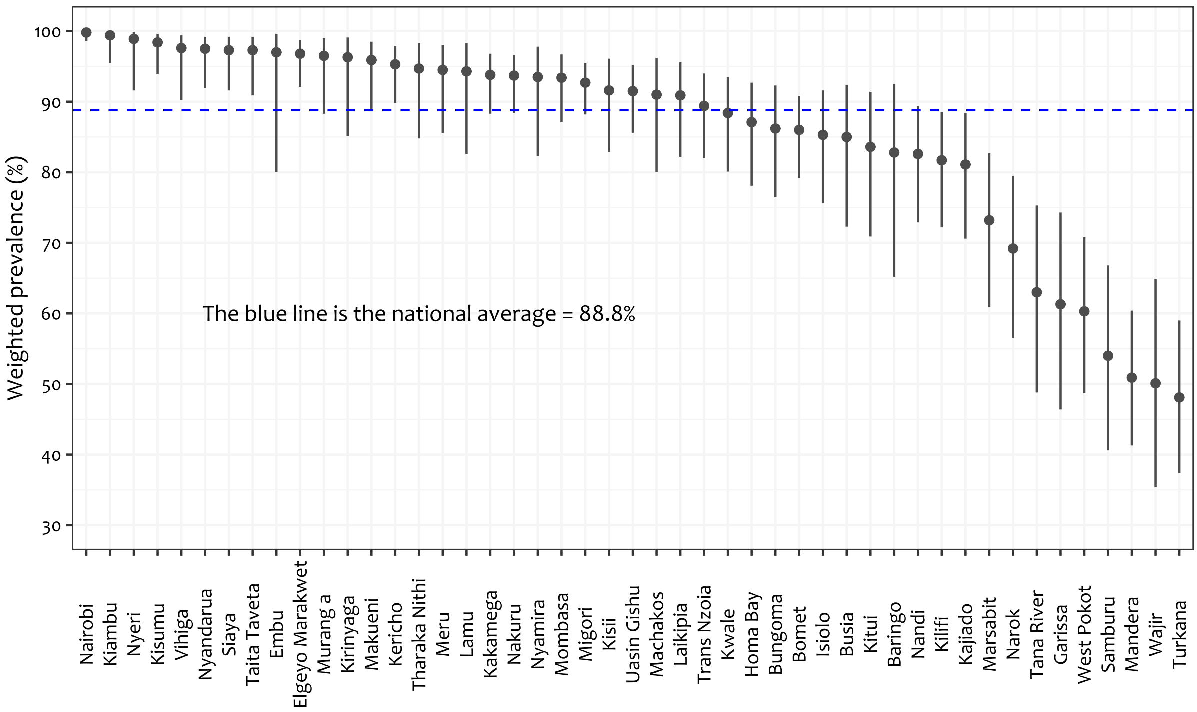

Figure 8: Weighted prevalence of skilled birth weight among caregivers of children 12-35 months at the subnationa levels. The blue line represents the national weighted prevalence

Spatial distribution of skilled birth attendance at the subnational level

```r
plot_map(
  merged_sba,
  fill = prevalence,
  breaks = c(40, 50, 70, 90, 100),
  labels = c("40-49.9", "50-69.9", "70-89.9", "90-100"),
  legend_title = "Prevalence (%)",
  subtitle = NULL,
  palette = 7,
  direction = -1) +
  theme(
    legend.text = element_text(size = 9),
    legend.background = element_blank(),
    legend.position = c(.18, .12),
```

```
    plot.background = element_rect(fill = "gray97"))

ggsave(
  "./Plots/Sba_map.png",
  width = 8,
  height = 6,
  dpi = 500,
  bg = "white")
```

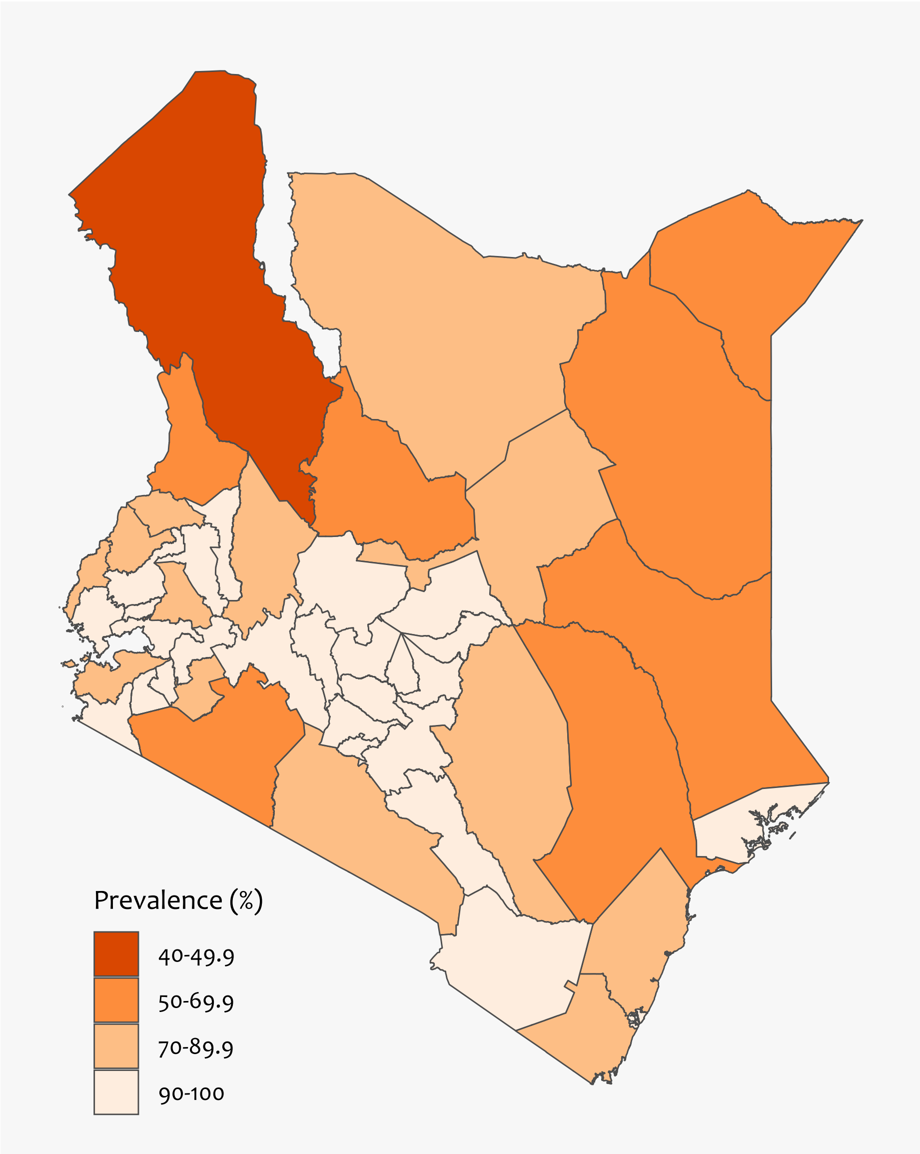

Figure 9: Spatial distribution of skilled birth attendance at the subnational level

Concentration curve for sba by cumulative percentage of the population ranked by wealth index

Data preparation step

```
cc_df_sba <- df_sba |>
    filter(!is.na(sba_num), !is.na(wealth_cont), !is.na(wt)) |>
    arrange(wealth_cont) |>
```

```
    mutate(
    pop_rank = cumsum(wt) - wt / 2, # midpoint rank within own weight mass
    pop_rank = pop_rank / sum(wt)) |>
  group_by(wealth_cont) |>   # collapse exact ties
  mutate(pop_rank = mean(pop_rank)) |> # give tied group its average rank
  ungroup() |>
  arrange(pop_rank) |>
  mutate(
        cum_prop = cumsum(wt) / sum(wt),
        cum_sba = cumsum(sba_num*wt) / sum(sba_num*wt))


cc_plot_data_sba <- bind_rows(
  data.frame(cum_prop = 0, cum_full_vacc = 0),
  cc_df_sba |> dplyr::select(cum_prop, cum_sba)) # Adding (0, 0) for plotting
```

Generating concentration index and plotting

```
## Concentration index
ci_result_sba <- ci(
  ineqvar = df_sba$wealth_cont,
  outcome = df_sba$sba_num,
  weights = df_sba$wt,
  type    = "CIw")

cc_plot_data_sba |>
  ggplot(aes(x = cum_prop, y = cum_sba)) +
  geom_line(color = "darkgreen", linewidth = 1) +
  geom_abline(
    slope = 1, intercept = 0,
     color = "#813134") +
  annotate(
    "text",
    x = 0.05, y = 0.95,
    label = paste0(
      "Concentration index (95% CI) = ",
      round(ci_result_sba$concentration_index, 2),
      " (",
      round(confint(ci_result_sba)[[1]], 2), ",", " ",
      round(confint(ci_result_sba)[[2]], 2), ")"),
    family = "Candara",
    hjust = 0, vjust = 1,
    size = 4.0) +
  labs(
    x = "Cumulative % of children, ranked by wealth (poorest \u2192 richest)",
    y = "Cumulative % of skilled birth attendance") +
  theme_classic(base_family = "Candara", base_size = 13) +
  theme(
    panel.grid = element_blank())
```

```
ggsave(
  "./Plots/Sba_concentration_curve.png",
  width = 7,
  height = 5,
  dpi = 500,
  bg = "white")
```

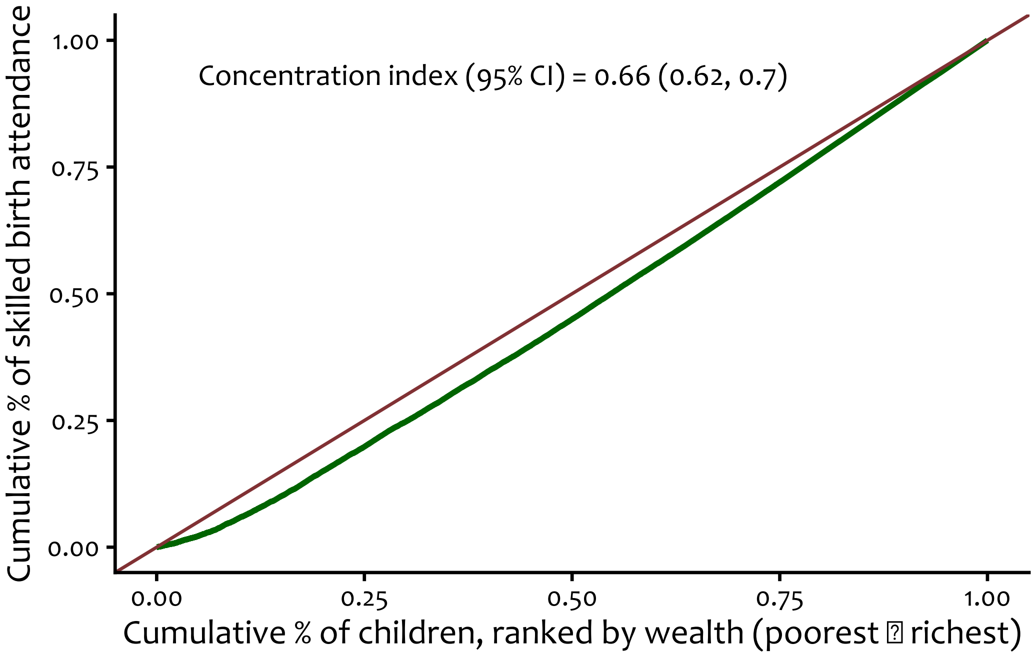

Figure 10: Concentration curves showing